

# 1 Similarity transformations of operators: computing $e^A B e^{-A}$

## 1.1 Motivation

Similarity transformations  $B \mapsto e^A B e^{-A}$  are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form: Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. If  $A^\dagger = -A$  (generator anti-Hermitian),  $U = e^A$  is unitary and the transformation preserves Hermiticity and spectra. This is the case in all examples below.

## 1.2 Method A: The Hadamard lemma

**Hadamard Lemma.** *For any two operators  $A, B$ ,*

$$e^A B e^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \text{ad}_A^n(B), \quad \text{with} \quad (1)$$

$$\text{ad}_A^n(B) \equiv [A, B]_n \equiv [A, [A, B]_{n-1}], \quad \text{ad}_A^0(B) \equiv [A, B]. \quad (2)$$

This series is especially helpful whenever the nested commutator yields zero at for a finite  $n$  or if the nested commutators eventually reproduce a small, recognizable pattern.

**Strategy 1: Reduce to elementary operators.** Utilize that for any  $A, X, Y$ ,

$$e^A (XY) e^{-A} = (e^A X e^{-A}) (e^A Y e^{-A}). \quad (3)$$

It thus suffices to compute  $e^A a e^{-A}$  and  $e^A a^\dagger e^{-A}$  once and any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

**Strategy 2: The dagger shortcut.** If  $A^\dagger = -A$ ,  $U = e^A$  is unitary with  $U^\dagger = e^{-A} = U^{-1}$ , so  $e^A B e^{-A} = U B U^\dagger$ . Therefore, taking the dagger simply yields

$$(e^A B e^{-A})^\dagger = e^A B^\dagger e^{-A}. \quad (4)$$

Once  $e^A a e^{-A}$  is known,  $e^A a^\dagger e^{-A}$  follows for free by Hermitian conjugation, roughly halving the bookkeeping.

## 1.3 Method B: The generator flow method (not proof read since not used yet)

Define  $X(\theta) \equiv e^{\theta A} X e^{-\theta A}$ ,  $X(0) = X$ . Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \text{ad}_A(X(\theta)). \quad (5)$$

This ordinary differential equation (ODE) is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around  $\theta = 0$  – but is can be more tractable in practice.

**Key fact: closure  $\Rightarrow$  finite-dimensional linear ODE.** If  $\text{ad}_A$  maps a finite set of operators  $\{X_1, \dots, X_n\}$  into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta} \vec{X}(\theta) = M \vec{X}(\theta), \quad [A, X_j] = \sum_i M_{ij} X_i, \quad \vec{X}(\theta) = e^{\theta M} \vec{X}(0), \quad (6)$$

reducing the problem to exponentiating an ordinary  $n \times n$  matrix.

## 1.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

## 1.5 Scope: when this machinery fails

The closure property requires  $A$  to be at most quadratic in a set of canonically (anti-) commuting operators. For a genuinely quartic operator such as  $H_{\text{int}} = U \sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$ ,  $\text{ad}_A(H_{\text{int}})$  does not close on a finite set for generic one-body  $A$  (e.g.  $A \propto J$  or  $A \propto T$ ) –  $[J, H_{\text{int}}]$  is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of  $A$ .

## 1.6 References

- J.J. Sakurai & J. Napolitano, *Modern Quantum Mechanics*.
- C. Gerry & P. Knight, *Introductory Quantum Optics*.
- D.F. Walls & G.J. Milburn, *Quantum Optics*.
- J.-P. Blaizot & G. Ripka, *Quantum Theory of Finite Systems*, MIT Press.
- A. Perelomov, *Generalized Coherent States and Their Applications*, Springer.
- J.W. Negele & H. Orland, *Quantum Many-Particle Systems*.
- B.C. Hall, *Lie Groups, Lie Algebras, and Representations*.